

Mathematics Extended Essay on the topic of Erdős' Probabilistic method

Author Name

May 15, 2026

Contents

1	Introduction	2
2	Scale Free Networks and Graph Theory	2
3	The Probabilistic Method	2
4	Predicting clique sizes in Scale Free Networks	2
5	Building Scale Free Networks	2
6	Comparing Results	2
7	Conclusion	2

1 Introduction

- A brief introduction - My motivation to explore this topic

2 Scale Free Networks and Graph Theory

- Describing the concept of graphs - Defining terms - Applying the idea to scale free networks

3 The Probabilistic Method

- Describing Probabilistic Notation - Applying using simple a simple Ramsey number example ($R(3,3) = 6$)

4 Predicting clique sizes in Scale Free Networks

- Setting out the problem: we want to find specific clique sizes in a scale free network with 4 hubs - Using the Chung-Lu Random Graph Model, we will describe how to statitcally generate a graph, with different probabilities of nodes and edges occuring. - Using the first moment we will explain the concept of

5 Building Scale Free Networks

6 Comparing Results

7 Conclusion